

SEHAT LEDGER

Zakat Capital Preserved Through Preventive Care

An offline-first clinical screening tool & outcome-gated impact ledger for the Ummah.

Presented by Active Bengaluru Foundation

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The Crisis: Leaking Zakat Capital

Late-Stage Care Consumes All

Community Zakat and Sadaqah funds are overwhelmed by late-stage diabetic and hypertensive complications.

- Dialysis: ■1,58,880 per patient/year
- Cardiovascular events & stroke
- Blindness & diabetic foot amputations

These are entirely preventable at a screening cost of just ■12–15.

The Zakat Preservation Math

Funding 1 Dialysis Patient for 1 Year
(■1,58,880 in community support)

EQUALS

10,600 to 13,200

Preventive Screenings

By identifying high-risk cases early and guiding them to primary care, we prevent organ failure and preserve community capital.

The Solution: Sehat Ledger PWA

1. Offline-First

Built for the field:

- Works in basement prayer halls with zero cellular signal.
- Local-first: writes data to localStorage immediately.
- Zero framework overhead: ~125 KB payload, loads natively.
- No app store blockages: installed directly via browser.

2. Vision Capture

Empowering volunteers:

- OCR via Gemini API reads glucometers & BP screens.
- Eliminates transcription & manual typing errors.
- Clean visual confirmation on a simple handset camera.
- Graceful fallback to manual when offline or keyless.

3. Secure Cloud Sync

Scale & Reliability:

- Direct PostgREST integration syncs to Supabase cloud.
- Conflict resolution using last-write-wins merging.
- Secure Row-Level Security (RLS) protecting patient records.
- Synchronizes silently when network status restores.

AI Powering the Care Loop

Structured AI Workflow

1. Vision API (Gemini-Flash)

Reads medical monitors. Refuses to guess rather than record bad clinical values.

2. Multilingual Referral (Translation)

Translates medical guidance into Urdu, Kannada, Hindi, or Tamil, tailored to patient.

3. Follow-Up Agent

Conversational AI that chats with patients, identifies barriers, and guides them to care.

Barrier Resolution Agent Demo

Patient barrier: "Clinic closed when I finish work"

Agent message:

"Assalamu Alaikum, we saw you had high blood pressure. Did you visit the Namma Clinic?"

Patient: "No, I finish work late at 7 PM."

Agent resolves: Suggests the Evening Clinic (open till 8 PM)

Instead of repeating the same referral, the agent addresses the specific barrier (distance, timing, cost) to close the loop.

The Ledger: Outcome-Gated Zakat Preservation

Strict Impact Attribution

We do NOT claim credit for:

- ✗ Screenings performed
- ✗ High-risk cases flagged
- ✗ Referral slips printed

We ONLY claim credit when:

- ✓ **Patient confirms they visited clinic**
- ✓ **First-line treatment is started**

Conservative Discounting:

*Every projection is discounted by 50%
to ensure the final numbers shown to the
mosque treasurer are completely defensible.*

Preserved Zakat Metrics

Accumulated deferred care burden:

■ **15,800 +**

Preserved Zakat Capital

Pathways Gated:

- Renal Failure avoided: ■ 3,575 / patient
- Cardiac Complications avoided: ■ 2,160
- Stroke Pathway avoided: ■ 1,980
- Diabetic Foot avoided: ■ 900
- Diabetic Vision loss avoided: ■ 1,260

*Attribution model uses local health-cost
parameters from ABF field studies in Bangalore.*

Sehat Ledger: The Vision

Low-Cost, High-Scale

Volunteers with ■7,000 Android phones become preventive health workers. No expensive devices or setups required.

Complete Closed Loop

AI does not just flag risk — it navigates patients around barriers, completes the referral, and validates the health outcome.

Measurable Zakat preservation

Turns mosque charity from late-stage bills to highly leveraged prevention.

Tech Highlights

- 100% Offline-First PWA
- Vanilla ES6 (No build step)
- 125 KB payload
- Supabase Sync via plain REST
- Gemini Vision & LLMs
- Privacy-First Design
- 57 Unit Tests (0 dependencies)

Try it online:

syedtousifmasood.github.io/Algorism.V2